

Supplementary Information

A spatially resolved hepatobiliary pharmacokinetic and pharmacodynamic framework for MASLD

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S1. Purpose and scope

This Supplementary Information records the model structure, state definitions, parameter values, scenario design, and code organization used for the manuscript. The implementation contains two related but distinct components. The coupled transport prototype represents a ten-position CV-to-PV reduced mesh with zoned transporter fields, metabolite handling, bile advection, obstruction, and an injury state. The gadoxetate scaffold is a separate three-compartment calibration model with plasma, hepatocyte, and bile amounts plus parallel renal clearance. The normalized transport prototype is not a clinical dosing model and the coefficient fields are hypothesis parameters pending functional calibration.

S2. Model architecture

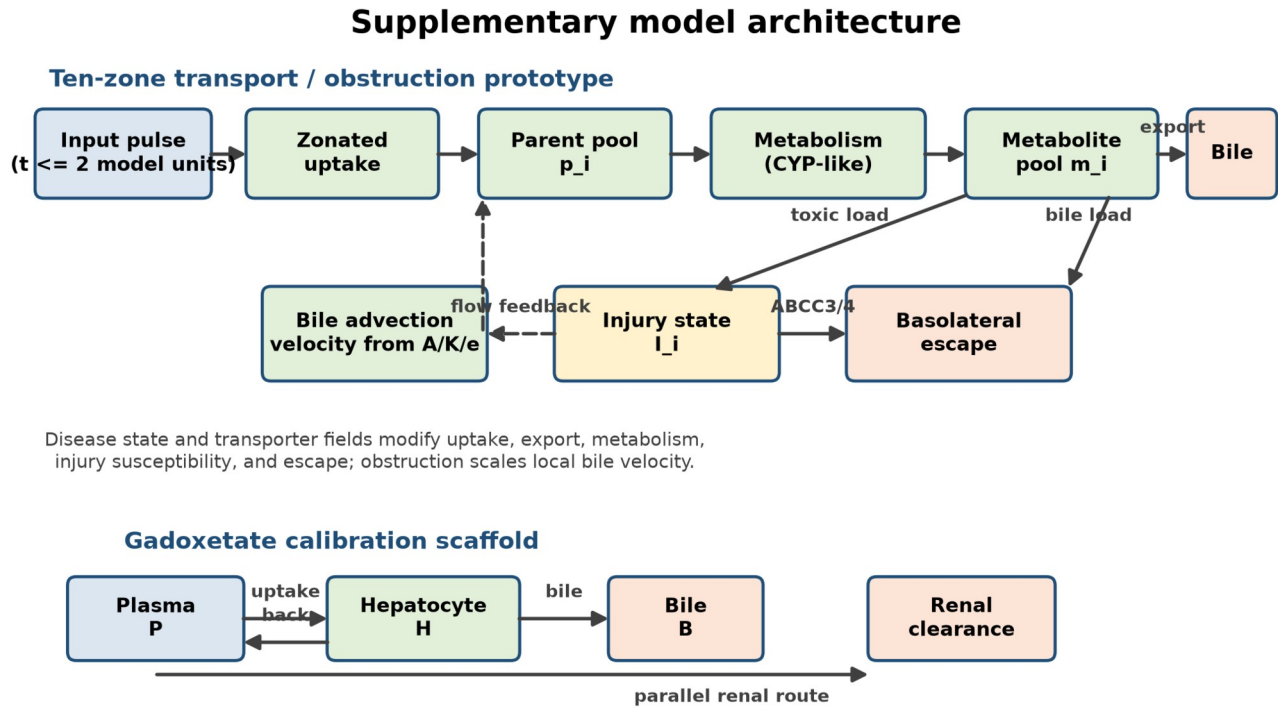


Figure S1. Computational architecture of the coupled transport prototype and the separate gadoxetate calibration scaffold.

S3. Coupled transport prototype

The prototype uses ten reduced positions ordered from central to portal: central, central, mid, mid, mid, periportal, periportal, portal, portal, portal. Michael Kuecken's normal-control and steatosis input profiles provide the local quantities A , K , epithelial volume fraction e , and connectivity. A normalized carrier velocity is calculated as $v_i = (A_i/K_i)e_i^{-1} c_i / \text{median}[(A/K)e^{-1} c]$. Obstruction scales this velocity by $(1 - \text{obstruction})$.

The implemented state vector is $y = [p_1 \dots p_{10}, m_1 \dots m_{10}, b_1 \dots b_{10}, I_1 \dots I_{10}]$, where p is the local parent pool, m is the metabolite/active hepatocyte pool, b is bile-side metabolite amount, and I is a normalized injury state. In this prototype, p_i is not a full plasma compartment; it is a zone-local parent pool driven by the input pulse.

S3.1 Governing equations

$$u_i = 0.10 * U_i * q_u * \exp(-1.7 i/n) * 1(t \leq 2)$$

$$\begin{aligned}
r_i &= 0.40 * CYP_i * p_i / (0.35 + p_i) \\
x_i &= f_m * C_i * X_i * 0.30 * m_i / (0.25 + m_i) \\
dp_i/dt &= u_i - r_i - 0.025 p_i \\
dm_i/dt &= r_i - x_i - 0.020 m_i \\
db_i/dt &= x_i/e_i + v_{(i-1)}b_{(i-1)} - v_i b_i \\
dL_i/dt &= 0.020 J_i L_i - 0.004 L_i + 0.003 E_i L_i
\end{aligned}$$

Here U_i is the aggregated uptake capacity, C_i is the aggregated canalicular capacity, E_i is the aggregated basolateral escape capacity, J_i is the disease injury factor, f_m is the MRP2 activity factor, q_u is the uptake inhibition factor, and X_i is the disease export factor. The toxic load is $L_i = \max(0, m_i - 0.05 \text{ GSH}_i) + 0.003 b_i$. Local flow is $\max(0.08, v_i(1 - 0.80 L_i))$. The first bile position receives an outflow term $-v_1 b_1$; upstream-to-downstream indexing follows the implementation's reduced CV-to-PV ordering.

S3.2 Fixed numerical settings

Item	Value	Interpretation
Time interval	0-60 model units	Transport prototype simulation window
Time grid	1201 points; delta t = 0.05	Output grid supplied to solve_ivp
Input pulse	1.0 for t <= 2.0; 0 otherwise	Dimensionless normalized input
Integrator	scipy.integrate.solve_ivp	Adaptive ODE integration
Relative tolerance	1×10^{-7}	Solver setting
Absolute tolerance	1×10^{-9}	Solver setting
Uptake inhibition	$q_u = 0.5$	50% activity condition
MRP2 activity	$f_m = 1.0$ or 0.35	Baseline or 65% inhibition
Obstruction	0.0, 0.5; LHS 0-0.75	Fractional velocity reduction

S4. Spatial transporter and disease fields

The four-zone fields are dimensionless relative capacities. They are explicit hypothesis parameters, not direct compound-specific functional measurements. The ten-position mesh expands these values using the zone labels listed above. Uptake is the mean of SLC10A1, SLCO1B1, and SLCO1B3; canalicular export is the mean of ABCB11, ABCC2, and ABCB4; basolateral escape is the mean of ABCC3 and ABCC4; ductular capacity is the mean of CFTR and SLC4A2.

State	Field	Portal	Periportal	Mid	Central
Control	Uptake	1.00	1.00	0.98	0.95
Control	Canalicular	1.00	1.00	1.00	1.00
Control	Escape	1.00	1.00	1.00	1.00
MASL	Uptake	0.98	0.96	0.92	0.88
MASL	Canalicular	1.00	0.97	0.92	0.88
MASL	Escape	1.05	1.08	1.12	1.16
MASH	Uptake	0.95	0.90	0.82	0.74
MASH	Canalicular	1.00	0.90	0.78	0.68
MASH	Escape	1.10	1.20	1.35	1.50

The individual gene-level fields, including ABCB11, ABCC2, ABCB4, CFTR, and SLC4A2, are preserved in `zonated_transport_genes.py`. The control, MASL, and MASH metabolism factors are respectively [1.00, 1.05, 1.00, 0.95], [0.95, 0.92, 0.88, 0.82], and [0.90, 0.85, 0.78, 0.68] from portal to central. Injury factors are [1.00, 1.00, 1.00, 1.00], [1.00, 1.05, 1.10, 1.15], and [1.05, 1.15, 1.30, 1.45].

S5. Michael input profiles

The full ten-row input arrays are embedded in `couple_michael_bileflow.py`. A, K, epithelial volume fraction e, and connectivity c are retained as source inputs for the normalized carrier field. The source file is the authoritative machine-readable parameter record; the summary below identifies the implementation and its use.

Input	Normal control	Steatosis	Use
Rows	10 CV-to-PV positions	10 CV-to-PV positions	Reduced spatial carrier
A	Source array in code	Source array in code	Conductance numerator
K	Source array in code	Source array in code	Conductance denominator
e	0.0310-0.0419	0.00287-0.0411	Velocity normalization

Connectivity c	0.7719-0.9824	0.0100-0.9436	Velocity weighting
Velocity scaling	median-normalized	median-normalized	Additional obstruction multiplier

S6. Gadoxetate calibration scaffold

The gadoxetate model is a separate three-state ODE system with a 24-hour simulation window. Amounts are in mmol for a 70 kg adult. The recommended dose is represented as 0.025 mmol/kg x 70 kg = 1.75 mmol. The label-inspired volume and clearance quantities are converted to an hour-based rate scale.

$$dP/dt = -(CL_{renal}/V_P)*renal*P - uptake*P + back*H$$

$$dH/dt = uptake*P - (back + bile)*H$$

$$dB/dt = bile*H$$

Scenario	uptake	back	bile	renal factor	Purpose
Healthy	1.10	0.12	0.112	1.00	Approximate 50:50 renal/hepatobiliary recovery
MASL/MASH exploratory	0.88	0.16	0.090	1.00	Reduced uptake and biliary handling
Severe hepatic impairment	0.35	0.24	0.025	1.30	Approximate 5% hepatobiliary recovery

Fixed quantity	Value
Body mass	70 kg
Dose	1.75 mmol
Vd approximation	0.21 L/kg x 70 kg = 14.7 L
Total clearance	0.250 L/min x 60 = 15.0 L/h
Renal clearance	0.120 L/min x 60 = 7.2 L/h
Hepatic remainder	7.8 L/h
Time grid	0-24 h; 2401 points

S7. Analysis design and endpoints

The counterfactual comparison holds the mean spatial capacity constant and compares zonated versus well-mixed fields. Primary quantities are bile AUC, hepatocyte parent AUC, bile Cmax, time to bile Cmax, peak injury, and terminal injury.

Analysis	Scenarios / ranges	Output
Counterfactual	Zonated vs well mixed; baseline, 50% uptake inhibition, 65% MRP2 inhibition	Relative bile and hepatocyte AUC differences
Scenario grid	MRP2 = 1.0 or 0.35; obstruction = 0 or 0.5	Bile Cmax, AUC, tmax, peak and terminal injury
LHS sensitivity	160 samples per normal-control/steatosis profile; MRP2 0.20-1.20; obstruction 0-0.75	Exposure and injury distributions
Gadoxetate	Healthy, exploratory MASL/MASH, severe impairment	24-hour route recovery and amount-time curves

S8. Code map and reproducibility

File	Role
couple_michael_bileflow.py	Defines input profiles, ten-position mapping, disease factors, ODE, obstruction, export, bile advection, and injury feedback.
zonated_transport_genes.py	Stores four-zone gene-level relative fields and aggregates uptake, canalicular, escape, and ductular capacities.
compare_zonated_wellmixed.py	Runs the mean-preserving spatial counterfactual and writes Figure 1 plus metric CSV.
run_jpkpd_analysis.py	Runs scenario grids and seeded Latin-hypercube sensitivity analysis; writes Figure 2 and CSV outputs.
simulate_gadoxetate_pk.py	Runs the separate plasma-hepatocyte-bile calibration scaffold and writes Figure 3 and CSV outputs.

create_second_paper_draft.py	Builds the main manuscript DOCX and inserts the three figures.
create_supplementary_information.py	Builds this supplementary document and its model architecture diagram.

From the project root, run `compare_zonated_wellmixed.py`, `run_jpkpd_analysis.py`, and `simulate_gadoxetate_pk.py` to regenerate numerical outputs and figures. Then run `create_second_paper_draft.py` and `create_supplementary_information.py` to regenerate the DOCX files. The sensitivity generator uses `numpy.random.default_rng(20260927)`.

The model is intentionally transparent rather than clinically validated. The next calibration release should replace relative transporter fields with zone-resolved protein or functional measurements, retain uncertainty distributions, and test predictions against independent gadoxetate and transporter-probe data.

S9. Relationship to the main manuscript

Figure S1 and the parameter tables provide implementation detail for the Methods and Results sections. The main manuscript reports normalized outputs and interprets them as structural-sensitivity results. No supplementary parameter should be interpreted as a clinical concentration, patient-specific coefficient, or validated toxicity threshold.